

06 — Implementation Rules

Version	0.1 (draft)
Date	2026-09-27
Based on	CLAUDE.md §3 and §5; BRD v0.6.1, PRD v0.6, HLD v0.2, LLD v0.1, Tasks v0.1 (all approved)
Status	AWAITING APPROVAL (one open decision, §4 D-10)

This document confirms **what we build with** and **the few places we deviate from CLAUDE.md §3**, with reasons. Everything not listed here follows CLAUDE.md §3 as written. Versions were checked against the npm registry on 2026-09-27. The lockfile (`pnpm-lock.yaml`) pins exact versions, and upgrades go through normal PRs with CI green.

1. Stack

1.1 Runtime and repository

Item	Choice	Version	Why
Language	TypeScript (strict mode)	6.0.x	One language for the UI and the 6 services. Not 7.0: typescript-eslint supports TypeScript < 6.1, and the lint and layer rules depend on it. We move to 7 when the linter supports it.
Runtime	Node.js LTS	24.x ("Krypton")	Current LTS, required by Vitest 5. The local machine has Node 20.13 and must be upgraded (§5).
Package manager	pnpm workspaces	12.x	Fast, strict dependency isolation per service
Monorepo build	Turborepo	2.11.x	Builds, tests and deploys only what changed. Each service stays independent (CLAUDE.md §3.9).

1.2 Front end (services/web)

Item	Choice	Version
Framework	Next.js (App Router) + React	16.3.x / 19.3.x
Styling	Tailwind CSS , with design tokens carried over from the approved prototype	4.3.x
Auth	Supabase Auth (Google provider) via <code>@supabase/supabase-js</code> + <code>@supabase/ssr</code>	2.117.x / 0.12.x
Streaming chat	Server-Sent Events from insight, rendered incrementally	—
UI tests	Playwright	1.63.x

1.3 Back-end services (intake, records, journeys, crm-engine, listings, insight)

Item	Choice	Version	Notes
HTTP layer	Hono , deployed as Vercel Node functions	4.13.x	One Hono app per service, built by the <code>libs/http</code> factory (task F-10)
Database access	Kysely (typed SQL query builder) + node-postgres (pg)	0.29.x / 8.23.x	No ORM: queries stay explicit so every one maps to an index in the LLD. Connects through the Supervisor pooler with prepared statements off.
Migrations	Plain SQL files in <code>services/<svc>/migrations</code> , run by the <code>libs/db</code> runner	—	Forward-only, expand → migrate → contract (CLAUDE.md §3.2)
Validation	Zod	4.6.x	Request and event validation at the edge
Contract types	openapi-typescript (types generated from <code>contracts/openapi</code>)	7.13.x	Drift check in CI (F-04)
JWT / JWKS	jose	6.2.x	Service tokens (R-2)
IDs	uuidv7	1.2.x	Conventions §2
Excel read/write	exceljs (streaming reader and writer)	4.4.x	Intake parsing and insight exports
PDF	@react-pdf/renderer	4.9.x	Proposal PDFs (journeys)
AI	@huggingface/inference	4.13.x	Hugging Face Inference Providers (pilot) → dedicated endpoint (paid). Always behind <code>libs/redaction</code> .
Supabase Storage	<code>@supabase/supabase-js</code> (Storage client only; the DB goes through Kysely)	2.117.x	Signed URLs

1.4 Data and platform (confirmed in HLD/LLD; restated)

Item	Choice
Database	Supabase PostgreSQL (Mumbai), a schema and roles per service, Supervisor pooling
Events and queues	Outbox + pgmq (ADR-0003)
Scheduler	pg_cron + pg_net
Files	Supabase Storage (private buckets; listings' public renditions bucket)
Hosting	Vercel , region <code>bom1</code> , 7 projects
Infrastructure as code	Terraform with the Vercel and Supabase providers (provider versions pinned in F-07)

1.5 Quality, testing and operations

Item	Choice	Version
Unit and integration tests	Vitest (integration against the Supabase local stack)	5.0.x
HTTP mocks in tests	MSW	2.15.x
Contract mocks for parallel work	Prism (from OpenAPI) + an event fixture publisher (from AsyncAPI)	5.16.x
Contract linting	Redocly CLI (OpenAPI), AsyncAPI CLI	2.54.x / 6.2.x
Lint	ESLint + typescript-eslint	10.11.x / 8.70.x

Item	Choice	Version
Layer enforcement	dependency-cruiser (domain <code>→/application/adapters</code> ; application <code>→/adapters</code>)	18.4.x
Secret scanning	gitleaks (GitHub Action + pre-commit)	pinned in F-02
Logging	pino (JSON) with a PII allow-list serializer	10.3.x
Tracing and metrics	OpenTelemetry (@opentelemetry/api 1.9.x, SDK 0.222.x, @vercel/otel 2.1.x)	as listed
Load tests	k6 (binary, pinned in REL-02)	—
CI/CD	GitHub Actions : one workflow per service (path filters), Vercel deploy on merge	—
Supabase / Vercel CLIs	supabase 2.118.x, vercel 60.x (devDependencies, run through pnpm)	—

2. Project conventions (in addition to docs/04-ld/conventions.md)

- Layout per service:** `src/domain` (pure TypeScript, with no imports of Hono, Kysely, pg, Supabase, Hugging Face or Node I/O), `src/application` (use cases and port interfaces), `src/adapters` (http, db, queue, storage, ai), and `src/main.ts` as the only composition root. `dependency-cruiser` enforces this in CI.
- `libs/` (infrastructure only, never domain models): `db`, `outbox`, `http`, `auth`, `observability`, `vocabulary`, `redaction`, `testing`. Each is versioned inside the workspace, and a change to a lib runs the CI of every service that uses it.
- Errors:** domain errors are typed classes mapped to RFC 7807 codes in the http adapter only.
- No PII in logs, traces, metric labels or event payloads.** The logger serializer drops non-allow-listed fields, and a test in every service asserts it.
- Every query is tenant-scoped** through the `libs/db` helper. A tenant-isolation test per service is mandatory (NFR-15).
- Money** is integer INR, **areas** are numbers in sq ft, **dates** are ISO 8601 (conventions §4).
- Commits:** Conventional Commits (the repo's commit-msg hook enforces them). One task = one PR = one branch `task/<ID>-<slug>`. The PR description links the task ID and ends with the attribution lines.
- Contract first:** a change to an API or event starts with the contract (`contracts/`, `tools/gen_events.py`). CI fails on drift. A breaking change needs a Change Request.

3. Environments and promotion

Environment	Now (Phase 1a pilot)	After the paid-plan gate
local	Supabase CLI (Docker via OrbStack) + <code>vercel dev</code> / <code>pnpm dev</code>	same
pilot (also dev)	Supabase Free <code>11e-crm-pilot</code> + Vercel Hobby, preview deployments per PR	becomes <code>dev</code>
staging	—	Supabase Pro (small) + Vercel Pro
production	—	Supabase Pro + PITR + Vercel Pro

The same build artifact (the Vercel deployment of a commit) is promoted between environments. Only environment variables differ.

4. Deviations from CLAUDE.md §3 (and §1 Stage 7)

#	CLAUDE.md rule	Deviation	Justification	Mitigation / end date
D-1	§3.8 Databases and internal services in private networks	Vercel ? Supabase and service ? service run over the public internet with TLS	Vercel has no private networking below Enterprise (Secure Compute). Product-owner hosting choice (H-1, H-7).	Short-lived service JWTs, SSL enforced, per-service DB roles, rotated secrets. Ends at the AWS move or with Vercel Secure Compute.
D-2	§3.8 Secrets only in a secrets manager	Vercel encrypted environment variables	No separate secrets manager on this platform in the pilot	Least scope per project, gitleaks in CI and pre-commit, 90-day rotation (data-hosting §3). Ends at the AWS move (Secrets Manager).
D-3	§3.6 Minimum 2 instances , autoscale on CPU/latency	Serverless functions: no fixed instances	Platform autoscales per request across zones	The DB pool caps act as the concurrency limit (capacity plan §2)
D-4	§3.9 and §4 Dockerfile per service	Vercel builds from source (<code>vercel.json</code> + Turborepo). No Dockerfile in Phase 1.	Vercel doesn't run containers	Each service still builds, tests and deploys alone. A Dockerfile per service is added as part of the AWS move.
D-5	§3.10 Four environments (local, dev, staging, production)	Phase 1a runs local + pilot only	Supabase Free allows 2 projects. The pilot uses anonymised data only (CR-005).	Staging and production are created at the paid-plan gate (REL-01)
D-6	§3.6 Sessions and locks in Redis / object storage / DB	No Redis: sessions via Supabase Auth, locks and rate limits in Postgres	The rule allows DB-backed state. Removes one vendor.	Revisit only if capacity triggers fire
D-7	§3.10 No manual console changes	A few settings the Terraform providers can't manage (e.g. Data API schema exposure, extension enabling) are applied by versioned scripts (<code>infra/scripts/</code>) or migrations, with a checklist in the runbook	Provider gaps	Every such setting is scripted or documented. None is set by hand without the script.
D-8	Versions (§1 Stage 6: "confirm versions")	TypeScript 6.0 instead of the newest 7.0	typescript-eslint peer range is < 6.1	Upgrade when supported
D-9	§3.5 Timeouts: default 2 s	Declared exceptions: intake <code>/v1/parse</code> 4 s; chat streaming (first token 3 s, total 15 s); PDFs and exports are async	Per NFR-7 and conventions R-5	Declared in <code>x-timeout-ms</code>
D-10	§1 Stage 7: one task at a time	OPEN, your decision: (a) keep one task at a time (≈ 185 working days to the pilot), or (b) run service tracks in parallel with Claude agents per service (≈ 49 – 69 working days to the pilot)	Contract mocks (F-05) make services independent. You still approve each service with <code>APPROVED: <service></code> , and each task still meets the full Definition of Done.	If (b): at most 3 service tracks in flight at once. The Foundation track is always done first and sequentially.

5. Local prerequisites (checked on this machine, 2026-09-27)

Tool	Needed	Found	Action
Node.js	24.x LTS	20.13.1	Upgrade (e.g. <code>fnm install 24</code> or <code>nvm</code>). Task F-01 adds <code>.nvmrc</code> / <code>engines</code> .
pnpm	12.x	not installed	<code>corepack enable</code> then <code>corepack prepare pnpm@12 --activate</code>
Docker	Any (for the Supabase local stack)	OrbStack	—
Supabase CLI	2.118.x	not installed	Installed as a devDependency by F-06
Vercel CLI	60.x	not installed	Installed as a devDependency by F-07
Terraform	1.x	not installed	Install before F-07 (e.g. <code>brew install terraform</code>)

Tool	Needed	Found	Action
Git + GitHub remote	—	No remote yet	Create a private GitHub repo before F-03 (CI needs it)

6. Definition of Done (every task in Stage 7)

From CLAUDE.md §5, with the project additions in *italics*:

- ☐ Matches the approved contract (OpenAPI / AsyncAPI) exactly. *CI drift check green.*
- ☐ Unit tests (domain), integration tests (adapters, against the Supabase local stack) and contract tests pass.
- ☐ Lint, type checks, import-layer checks (dependency-cruiser) and secret scan (gitleaks) pass.
- ☐ Logs, traces, metrics and health checks in place. *The PII-not-logged test passes.*
- ☐ *The tenant-isolation test passes (every service that touches tenant data).*
- ☐ Migrations are backward compatible (expand → migrate → contract).
- ☐ Service README updated (run locally, env vars, owned data, events).
- ☐ `docs/STATUS.md` updated with the task ID marked done.
- ☐ *The PR follows Conventional Commits and links the task ID.*